

Making a Diagnosis of Epilepsy

Epilepsy is a disorder of recurrent unprovoked (or reflex) seizures. The key to the diagnosis of epilepsy is estimating the risk of seizure recurrence.

It may be diagnosed if at least two unprovoked (or reflex) seizures occur at least 24 hours apart. Studies have established that the risk of seizure recurrence after two unprovoked seizures is greater than 50% over the next 10 years, it is even higher after three unprovoked seizures.

However, epilepsy may also be diagnosed after a first-time seizure if the clinician estimates that the risk of future recurrences is high (similar to that after at least two seizures) or if there is evidence of an underlying epilepsy syndrome [1].

A history of prior brain insult, epileptogenic abnormalities on EEG, significant brain imaging abnormalities or nocturnal seizures confer an increased risk of future recurrence after a first-time seizure; enabling a diagnosis of epilepsy in the appropriate clinical context [2].

Conversely, epilepsy may be considered “resolved” for those who have outgrown an age-dependent epilepsy syndrome (Chapter 21 describes epilepsy syndromes) and those who have been seizure-free for at least 10 years with the last 5 years off antiseizure medications [1,2].

Describing the Epilepsy Disorder

Once you have diagnosed a patient with epilepsy, the next step is to classify it. The following approach is based on the ILAE classification of epilepsies [3]:

1. The first step is to diagnose the prevalent seizure type(s) as described in Chapter 19.
2. Next, diagnose the type of epilepsy based on the prevalent seizure type(s), comorbidities, and ancillary findings such as EEG (interictal and ictal) and neuroimaging abnormalities.
3. Focal onset seizures suggest focal epilepsy, generalized onset seizures suggest a generalized epilepsy and the presence of both seizure types suggests a mixed epilepsy.

4. Once you have determined the type of epilepsy (focal, generalized, or mixed), then consider if it fits into an epilepsy syndrome (often age specific). Focal epilepsies may or may not be syndromic, while generalized epilepsies are usually syndromic. Epilepsy syndromes are discussed in Chapter 21.
5. Finally, whether or not the epilepsy is syndromic, it is important to consider potential underlying etiologies.
6. Etiologically, epilepsies may be due to a structural (e.g., stroke) or genetic (e.g., SCN1A mutation with Dravet syndrome), infectious (e.g., neurocysticercosis), immune (e.g., anti-NMDA receptor encephalitis), or metabolic causes (e.g., pyridoxine dependent seizures). Not infrequently, the etiology is unknown.

With these steps, you will be able to comprehensively describe the epilepsy disorder.

Types of Epilepsies

Focal Epilepsy

Focal epilepsies are characterized by focal onset seizures types. Focal interictal and ictal EEG abnormalities (slowing or discharges) and neuroimaging findings (lesions) are supportive. Some focal epilepsies are syndromic. Focal epilepsies are subclassified as unifocal, hemispheric, or multifocal, they may also be grouped based on lobar involvement.

Frontal Lobe Epilepsy: Frontal lobe epilepsy is typically characterized by motor seizures (hyperkinetic or asymmetric tonic posturing), awareness may or may not be impaired. These seizures are typically nocturnal (arising out of non-rapid eye movement sleep) and often cluster. The interictal EEG is often normal though repeated EEGs may show epileptogenic discharges which may appear bifrontal or midline. The ictal EEG is often difficult to interpret, it may be normal or completely obscured by movement artifact during the seizure. Less commonly, a focal frontal ictal pattern may occur. Ictal patterns may be preceded by focal or diffuse suppression and/or low-voltage fast activity. False localization of the ictal pattern to the ipsilateral temporal lobe or false generalization (secondary bilateral synchronization) may also occur [4]. Figure 20.1 shows a frontal lobe seizure.

Temporal Lobe Epilepsy: Temporal lobe epilepsy is typically characterized by focal onset impaired awareness and nonmotor seizures (behavioral arrest). Temporal seizures may originate from the mesial or lateral parts of the temporal lobe as described:

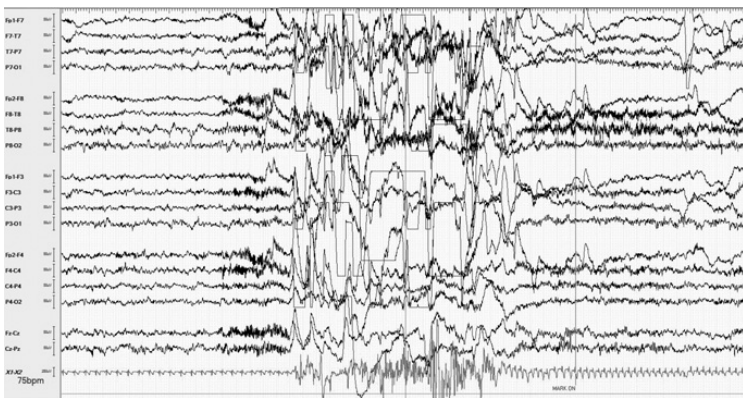


Figure 20.1 Seventeen-year-old girl with frontal lobe epilepsy, frontal-predominant high-amplitude discharges (midline channels), and movement artifact during nocturnal seizure (time base 60 s).

1. Mesial temporal lobe epilepsy typically has autonomic (rising epigastric sensation/abdominal discomfort), sensory (olfactory/gustatory), cognitive (*déjà vu*/jamais vu), or emotional (fear) onset seizures. Impaired awareness, behavioral arrest, and oral/manual automatisms are common. Focal to bilateral tonic clonic evolution (secondary generalization), if it occurs typically late in the seizure. The interictal EEG may show slowing (temporal intermittent rhythmic delta activity) and/or discharges in the anterior temporal channels (such as F7/F8). The ictal EEG is characterized by rhythmic alpha-theta or spikes. Postictal focal slowing (ipsilateral to the focus) is common. Figure 20.2 shows temporal intermittent rhythmic delta activity (TIRDA) while Figure 20.3 shows a mesial temporal lobe seizure.
2. Lateral temporal lobe epilepsy (LTLE, also called neocortical epilepsy) typically has auditory or vertiginous aura that precedes focal onset seizures with impaired awareness and behavioral arrest. Secondary generalization is more frequent and rapid compared to mesial temporal lobe seizures. The interictal EEG may show slowing and/or discharges in the mid-temporal or posterior channels (such as T7/T8). The ictal EEG shows rhythmic patterns with variable amplitudes and frequencies associated with rapid spread [5]. Figure 20.4 shows a lateral temporal lobe seizure.

Parietal Lobe Epilepsy: Parietal lobe epilepsy is typically characterized by focal nonmotor onset seizures which may be positive (paresthesia)/negative(numbsness) sensory phenomena. Disorientation, complex visual hallucinations, vertiginous, and visual illusions or disturbances of body

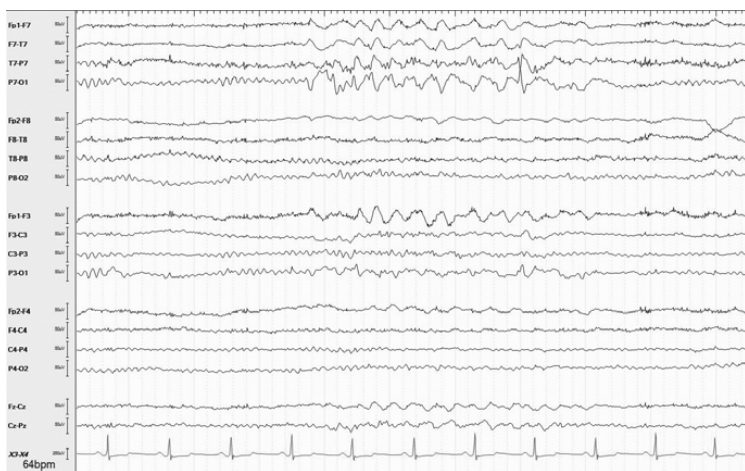


Figure 20.2 Fifty-year-old woman with mesial temporal sclerosis, left temporal intermittent rhythmic delta activity (TIRDA).

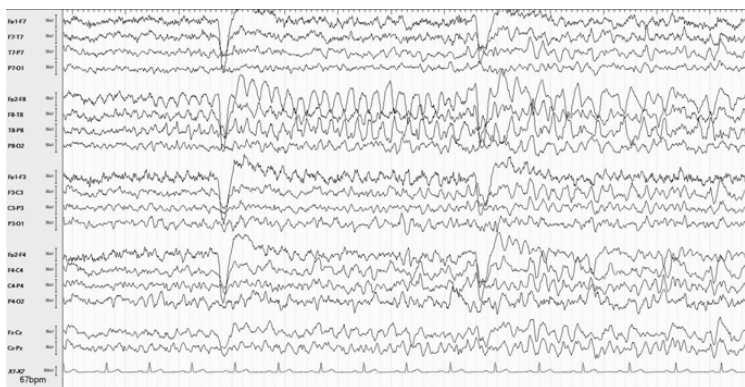


Figure 20.3 Twenty-three-year-old woman with mesial temporal lobe epilepsy, right anterior temporal sharp rhythmic theta activity at ictal onset of seizure.

image may occur along with receptive aphasia and ipsilateral or contralateral rotatory body movements. The interictal EEG may show abnormalities in the posterior channels, these can be precipitated by

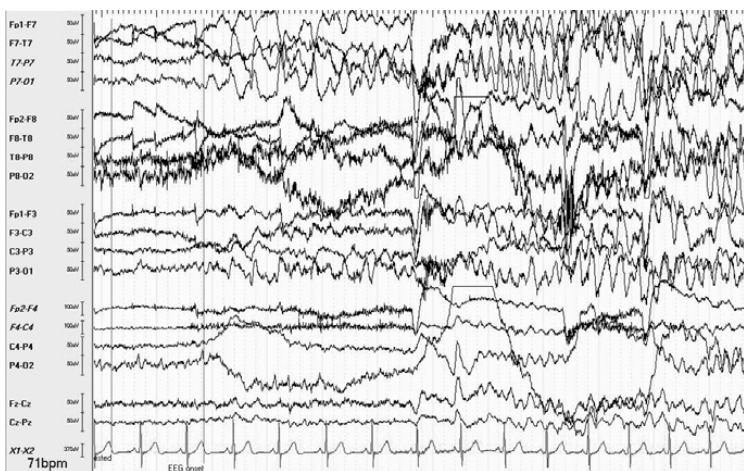


Figure 20.4 Twenty-eight-year-old woman with lateral temporal lobe epilepsy, left temporal sharp rhythmic delta-theta activity at ictal onset with rapid bilateral spread.

reflex sensory stimulation/touch. The ictal EEG is often unhelpful being normal or shows nonspecific slowing [6].

Occipital Lobe Epilepsy: Occipital lobe epilepsy is typically characterized by focal nonmotor onset seizures, which are often simple visual phenomena. Oculomotor features such as forced eye closure, eyelid flutter, eye deviation, and nystagmus also occur. The interictal EEG may show an asymmetric posterior-dominant alpha or photic responses and high-amplitude occipital spikes (Panayiotopoulos syndrome). The ictal EEG is characterized by occipital paroxysmal fast or rhythmic spike activity; lateralized or generalized ictal rhythms may also occur [7]. Figure 20.5 shows an occipital seizure.

Insular Epilepsy: The insula is an in-folded region of cortex located within the lateral sulcus that is sometimes referred to as a fifth lobe. Insular seizures are typically characterized by cervical-laryngeal discomfort (choking), dyspnea, painful paresthesia, or temperature alterations, though they can mimic frontal, temporal, or parietal seizures. Interictal and ictal surface EEG is seldom useful given the deep location of seizure foci in this form of epilepsy [8].



Figure 20.5 Nineteen-year-old woman with mitochondrial dysfunction, left occipital seizure (O1) with bilateral spread.

Generalized Epilepsy: Generalized epilepsies are characterized by generalized onset seizures types. Generalized ictal and interictal EEG abnormalities (generalized spike waves) are supportive, as is a positive family history. Generalized epilepsies are considered to have a genetic etiology though they aren't always inherited and may result from spontaneous mutations. Some authors use the term genetic generalized epilepsy in cases with strong inheritance and idiopathic generalized epilepsy (IGE) for those without it, however, these terms are used interchangeably. Generalized epilepsies are usually syndromic such as childhood absence epilepsy (CAE), juvenile absence epilepsy (JAE), or juvenile myoclonic epilepsy (JME).

Multifocal and Mixed Epilepsy (Focal and Generalized Epilepsy)

Multifocal epilepsies typically result from diffuse and symptomatic brain injuries such as global ischemia, encephalitis, or trauma. They are associated with multifocal discharges. Figure 20.6 shows multifocal discharges in a child with cerebral palsy.

Mixed epilepsies are characterized by both focal and generalized onset seizures and similar ictal and interictal EEG abnormalities. There are usually syndromic such as Dravet syndrome and Lennox–Gastaut syndrome. Common epilepsy syndromes are described in the following chapter (Chapter 21).

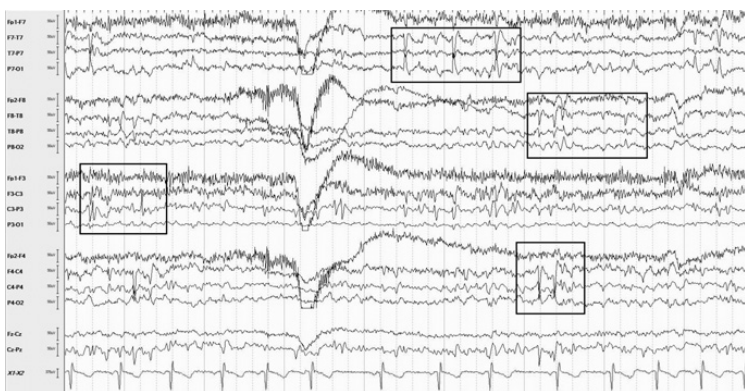


Figure 20.6 Six-year-old boy with cerebral palsy, multifocal spikes involving multiple channels (T7, P7, T8, C3, and C4).

Chapter Summary

1. Epilepsy is a disorder of recurrent unprovoked (or reflex) seizures; the key to diagnosing epilepsy is estimating the risk of recurrence.
2. Epilepsy may be diagnosed if a patient has two or more seizures at least 24 hours apart, a first-time seizure with factors that increase the risk of recurrence or an epilepsy syndrome.
3. When determining the type of epilepsy, first identify the seizure type(s), then the corresponding epilepsy type.
4. Consider if the epilepsy is potentially syndromic or nonsyndromic.
5. Consider the underlying etiology (structural, genetic, infectious, immune, or metabolic).
6. Focal epilepsies may be classified based on their region of onset, that is, frontal, temporal (mesial/lateral), parietal, occipital, and insular.
7. Ictal and interictal EEG findings vary with the type of focal epilepsy.

References

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